

Zacharia Hammad

Software Engineer in AI Products and Developer Tools

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Software engineer building AI products and developer tools in Python, TypeScript, and Rust. Owns computer-vision development from data and model training through user-facing applications.

PROFESSIONAL EXPERIENCE

Visionary Solutions / Software Engineer

Aug 2025 - Present

- Owned a computer-vision product from data sourcing and labeling through model training and the user-facing application.
- Built shared TypeScript voice-agent tool execution across VAPI and OpenAI Realtime, binding actions to verified caller identity.
- Built Neo4j infrastructure for identity matching and conversation context, with access controls for AI applications.
- Built YOLO/ONNX training and release tooling with MLflow tracking, evaluation, and model promotion/rollback; integrated NVIDIA DeepStream inference.
- Built reliable event delivery between services using PostgreSQL and Kafka.

PECO / Exelon / UAS Engineering Co-op

Sep 2023 - Mar 2024

- Automated drone reporting with Python/APIs, cutting preparation from two hours to under 30 minutes.

NAVSEA / Engineering Co-op

Sep 2022 - Mar 2023

- Improved submarine simulation algorithms and sensor models; documented gas-sensor workflows.

Saint-Gobain / Research and Development Co-op

Sep 2021 - Mar 2022

- Analyzed materials experiments with Python, MATLAB, and statistics to identify process improvements.

SELECTED PROJECTS

RepoToire / Rust and Quint

- Built a Rust runtime prototype for AI coding agents with validated execution plans, staged execution, bounded retries, and Quint-to-Rust conformance tests.

ML Pothole Detection System / Drexel Senior Design

- Developed YOLO edge inference with GPS-tagged capture, backend APIs, PostgreSQL, and a geospatial dashboard for a Drexel Senior Design Championship-winning team project.

ASIC Design Optimization / Drexel Course Project

- In a two-person project, adapted Tcl scripts for Synopsys design flows to compare timing and area across five benchmark circuits using SkyWater and ASAP7 libraries.

TECHNICAL SKILLS

Languages Python, TypeScript, JavaScript, Rust, SQL, Tcl

Product and APIs React, Next.js, FastAPI, LLM APIs, VAPI, OpenAI Realtime

Data and ML PostgreSQL, Neo4j, Kafka, YOLO, ONNX, NVIDIA DeepStream, MLflow

EDUCATION AND CERTIFICATION

Drexel University

Jun 2025

B.S. Computer Engineering | Minor in Data Science | GPA 3.69

Neo4j Certified Professional | Aug 2026